FLEET TEAM DIGEST

737MAX-FTD-32-17024

Issue Title : Main Gear Brakes Leakage Limits

Airplane Model	ATA	Minor Model(s)
737MAX	3246-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG, 757, 767, 737, 727, 747, 777	9E991

Originated Date	Last Revised Date	Created On
05/11/2017	05/11/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All airplane sub-models

Description

Boeing was asked to provide explicit hydraulic fluid leakage limits relative to main gear brakes. Chapter 29 defines hydraulic fluid external leakage “Normal Operation Limits” as well as “Dispatch Limits To Avoid Delay” in terms of drops/minute for various hydraulic components. However, main gear brakes are not one of the explicitly identified components.

Background

In the past, brake leakage limits were covered by the limits provided in Chapter 29 for static and dynamic seals.

However, these general seal leakage limits are considered too high for brake leakage. Ideally, landing gear brakes should be removed from service if there is any leakage. Brakes do not have an acceptable “normal operation” leakage rate. If the “normal operation” limits for general static or dynamic seals defined in Chapter 29 are applied to brakes, a significant amount of hydraulic fluid could be deposited on the heat sink of the brake or surrounding structure during a long flight. While brakes are designed to be tolerant of a variety of contaminants, including hydraulic fluid, significant exposure could affect stopping performance and brake stability, and increase the risk of brake fires. Hydraulic fluid contamination is particularly a concern for brakes utilizing a carbon heat sink material.

Applying a zero-leakage limit alleviates the contamination concerns. However, this is extremely burdensome for operators to comply with, and would likely result in unnecessary AOG situations, particularly for airplane models that cannot dispatch with a brake deactivated. Brakes exhibiting very minor leakage (i.e. wetness or seepage) can be put into service with no negative consequences provided the condition is monitored regularly for escalation and is corrected at the next convenient opportunity.

Status

closed

Interim Action

N/A

Final Action

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A specific leakage check procedure and applicable leakage limits for main gear brakes are located in AMM 32-41-41/601.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	32-41-41/601		

Related Categories

- In Service Occurrences : null

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